Capstone Project:Flipkart E-Commerce Product

PHASE 2: Data Preprocessing,EDA and Feature Engineering

Step 1: Import Libraries & Load Data

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

In [27]:
```
df = pd.read_csv("Flipkart_scrap.csv")
df.head()
```

Out[27]:

	Unnamed: 0	name	price	original_price	reviews	Rating	Bank_offers
0	0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	NaN	NaN	NaN
1	1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	₹265	₹1,299	(923)	3.5	Super Deals
2	2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	₹327	₹2,999	(339)	4.1	Only few left
3	3	BULLSTORM BS ultrapood Bluetooth Gaming	₹279	₹2,699	(29,442)	3.4	Bank Offer
4	4	TECHIO Black Earbuds | True Wireless Stereo | ...	₹235	₹1,980	(11,657)	3.6	Only few left

Step 2: Basic Data Understanding

In [28]:
```
df.shape
```

Out[28]:
```
(1000, 7)
```

In [29]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Unnamed: 0      1000 non-null   int64
 1   name            1000 non-null   object
 2   price           997 non-null    object
 3   original_price  991 non-null    object
 4   reviews         912 non-null    object
 5   Rating          912 non-null    float64
 6   Bank_offers     922 non-null    object
dtypes: float64(1), int64(1), object(5)
memory usage: 54.8+ KB
```

In [30]:
```
df.describe()
```

Out[30]:

	Unnamed: 0	Rating
count	1000.000000	912.000000
mean	499.500000	3.936513
std	288.819436	0.266650
min	0.000000	2.800000
25%	249.750000	3.700000
50%	499.500000	4.000000
75%	749.250000	4.100000
max	999.000000	5.000000

In [31]:
```
df.isnull().sum()
```

Out[31]:
```
Unnamed: 0         0
name               0
price              3
original_price     9
reviews           88
Rating            88
Bank_offers       78
dtype: int64
```

Step 3: **Data Cleaning & Preprocessing**

Drop unwanted column

In [32]:
```
# Check if the column exists before dropping it
if "Unnamed: 0" in df.columns:
    df.drop(columns=["Unnamed: 0"], inplace=True)
else:
    print("Column 'Unnamed: 0' not found in the DataFrame")
```

In [33]:
```
df.head()
```

Out[33]:

	name	price	original_price	reviews	Rating	Bank_offers
0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	NaN	NaN	NaN
1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	₹265	₹1,299	(923)	3.5	Super Deals
2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	₹327	₹2,999	(339)	4.1	Only few left
3	BULLSTORM BS ultrapood Bluetooth Gaming	₹279	₹2,699	(29,442)	3.4	Bank Offer
4	TECHIO Black Earbuds | True Wireless Stereo | ...	₹235	₹1,980	(11,657)	3.6	Only few left

Clean price columns (remove ₹ and comma)

In [34]:
```
# First check the data type of the columns
df["price"] = df["price"].astype(str).str.replace("₹","", regex=False).str.replace(",","", regex=False).astype(float)
df["original_price"] = df["original_price"].astype(str).str.replace("₹","", regex=False).str.replace(",","", regex=False).astype(float)

# The corrected code:
# 1. Converts the columns to string type first using astype(str)
```

```
# 2. Uses str accessor before replace() to properly access string methods
# 3. Removes the ₹ symbol
# 4. Removes commas
# 5. Converts to float
# 6. Uses regex=False for better performance
```

Clean reviews column

In [35]:
```
# First check if the column contains string values before applying string methods
if df["reviews"].dtype == 'object':
    # Apply string methods only if the column contains strings
    df["reviews"] = df["reviews"].str.replace("(","").str.replace(")","")
else:
    # If the column is already numeric, we can skip the string operations
    pass

# Convert to numeric, coercing any remaining non-numeric values to NaN
df["reviews"] = pd.to_numeric(df["reviews"], errors="coerce")
```

Handle missing values

In [36]:
```
df["Rating"].fillna(df["Rating"].median(), inplace=True)
df["reviews"].fillna(df["reviews"].median(), inplace=True)
df["Bank_offers"].fillna("No Offer", inplace=True)
```

Step 4: **Feature Engineering**

Discount percentage

In [37]:
```
# Convert price columns from strings to numeric values
df["original_price"] = pd.to_numeric(df["original_price"], errors='coerce')
df["price"] = pd.to_numeric(df["price"], errors='coerce')

# Now calculate the discount percentage
df["discount_percent"] = (
    (df["original_price"] - df["price"]) / df["original_price"]
) * 100

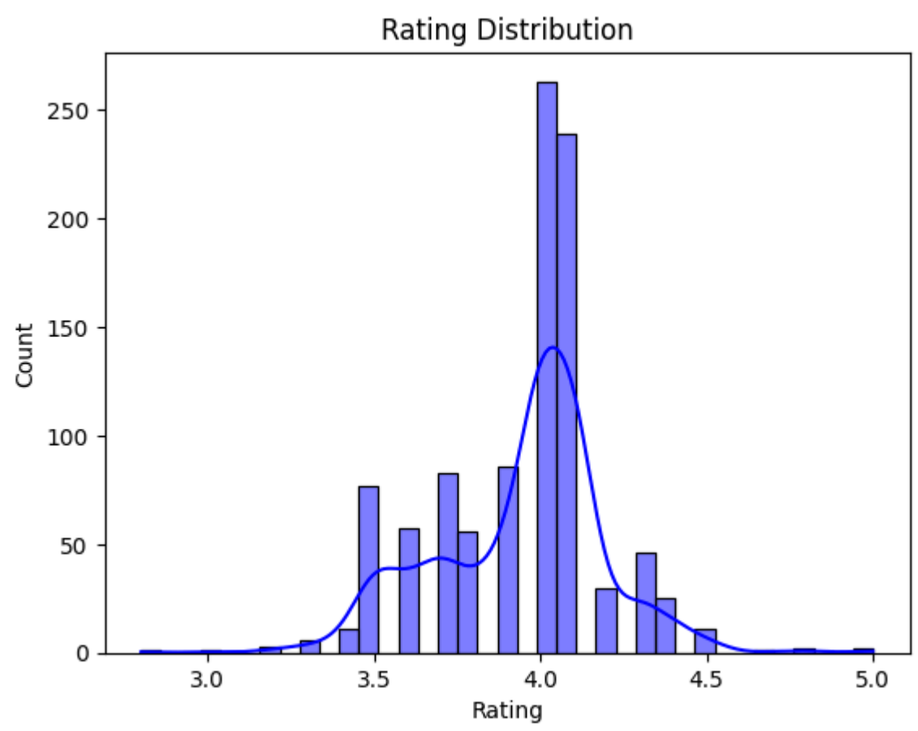
# Optional: Handle any NaN values that might result from the conversion
# df["discount_percent"] = df["discount_percent"].fillna(0)
```

In [38]:
```
df["discount_percent"].fillna(df["discount_percent"].median(), inplace=True)
```

Step 5: **Exploratory Data Analysis (EDA)**

Rating distribution

In [39]:
```
plt.figure()
sns.histplot(df["Rating"], kde=True,color="blue")
plt.title("Rating Distribution")
plt.show()
```

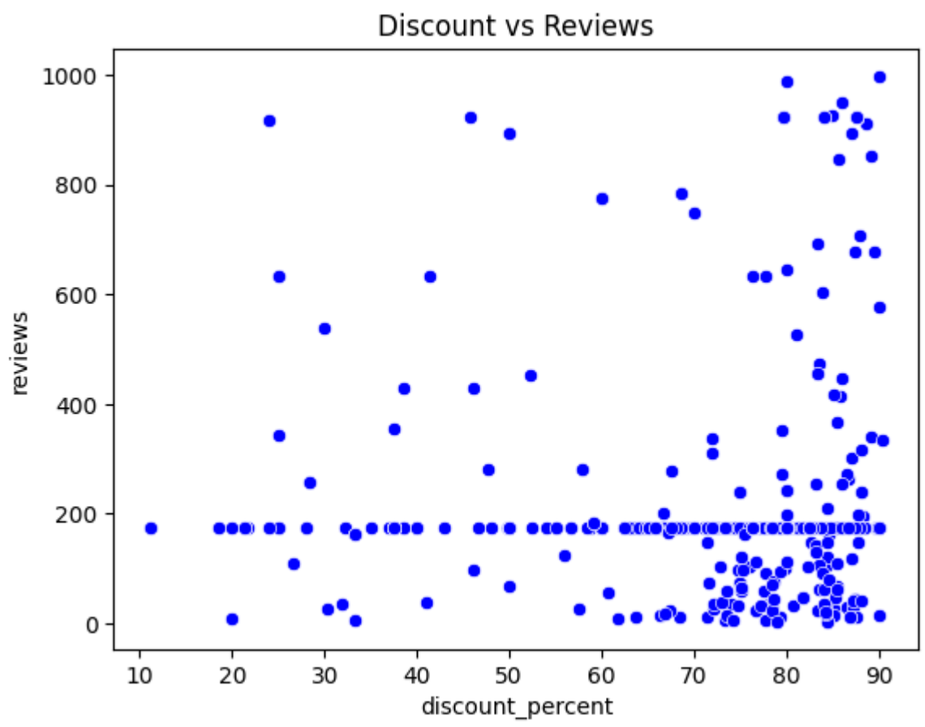


Price vs Rating

In [50]:
```
plt.figure()
sns.scatterplot(x="price", y="Rating", data=df,color="coral")
plt.title("Price vs Rating")
plt.show()
```

Discount vs Reviews

In [51]:
```
plt.figure()
sns.scatterplot(x="discount_percent", y="reviews", data=df,color="blue")
plt.title("Discount vs Reviews")
plt.show()
```



Correlation Heatmap

In [42]:
```
print(df.columns)
```

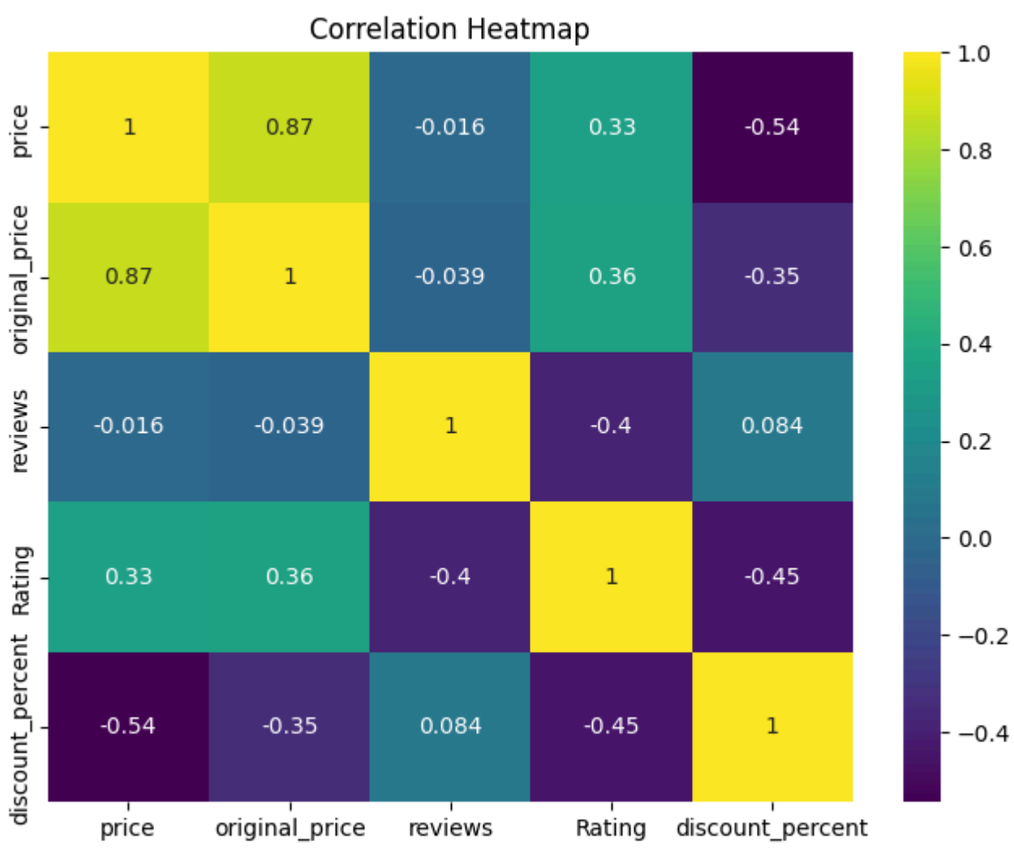
```
Index(['name', 'price', 'original_price', 'reviews', 'Rating', 'Bank_offers',
       'discount_percent'],
      dtype='object')
```

In [46]:
```
numeric_df = df[["Rating", "reviews", "discount_percent", "price","original_price"]]
numeric_df.corr()
```

Out[46]:

	Rating	reviews	discount_percent	price	original_price
Rating	1.000000	-0.398519	-0.446317	0.328740	0.356218
reviews	-0.398519	1.000000	0.083804	-0.015694	-0.038906
discount_percent	-0.446317	0.083804	1.000000	-0.544008	-0.346988
price	0.328740	-0.015694	-0.544008	1.000000	0.872677
original_price	0.356218	-0.038906	-0.346988	0.872677	1.000000

In [48]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="viridis")
plt.title("Correlation Heatmap")
plt.show()
```



Top 10 Most Reviewed Products

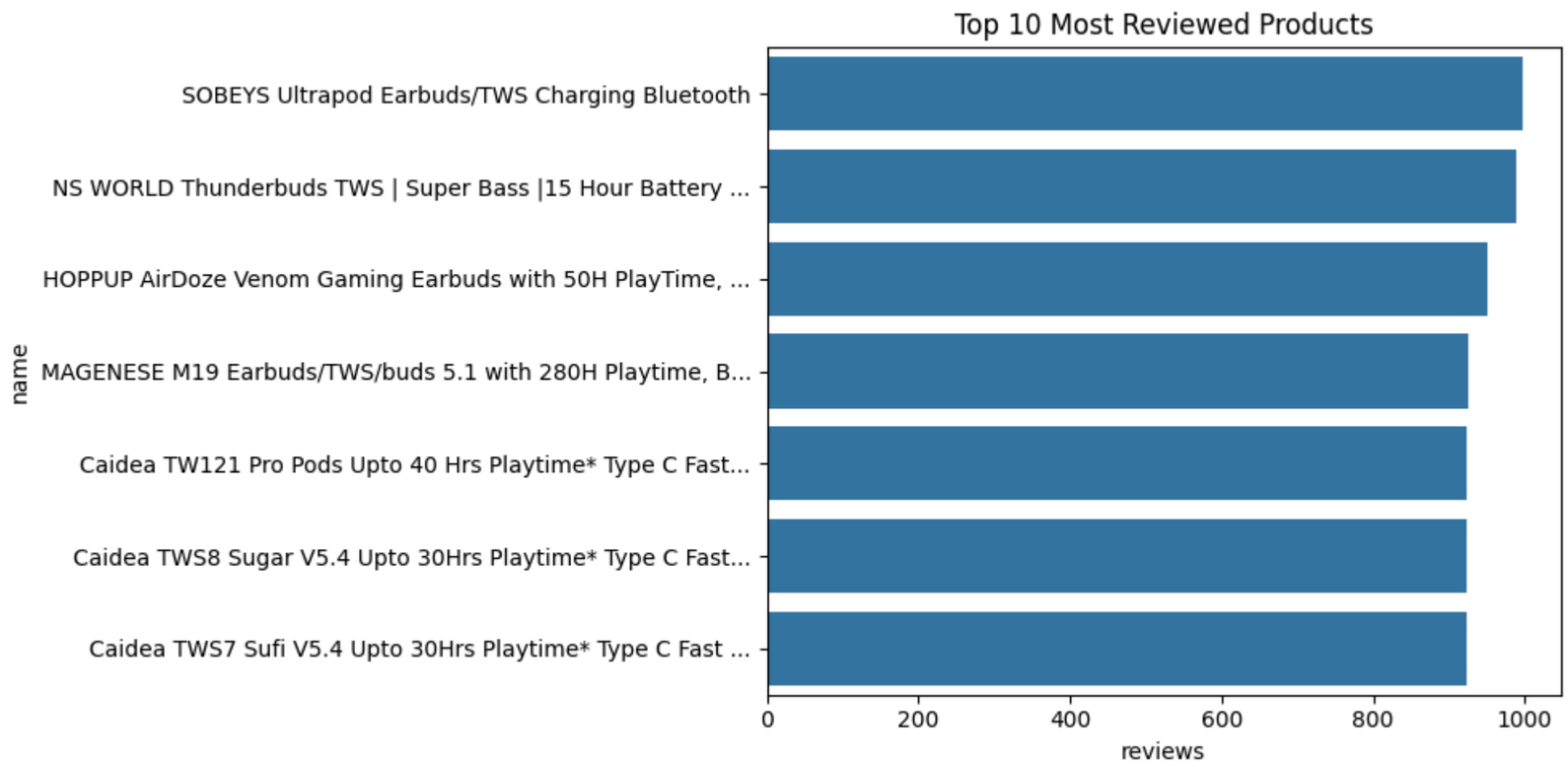
In [49]:
```
# First, let's check what columns are actually available in the DataFrame
print("Available columns:", df.columns.tolist())

# Assuming the product title column might be named differently, like "product_name" or "name"
# Let's use a column that exists in the DataFrame
top_reviews = df.sort_values("reviews", ascending=False).head(10)

plt.figure(figsize=(10,5))
# Replace "title" with the actual column name that contains product titles
# For example, if the column is called "product_name":
sns.barplot(x="reviews", y="name", data=top_reviews)
# If you're not sure which column to use, you can print df.head() to inspect the DataFrame

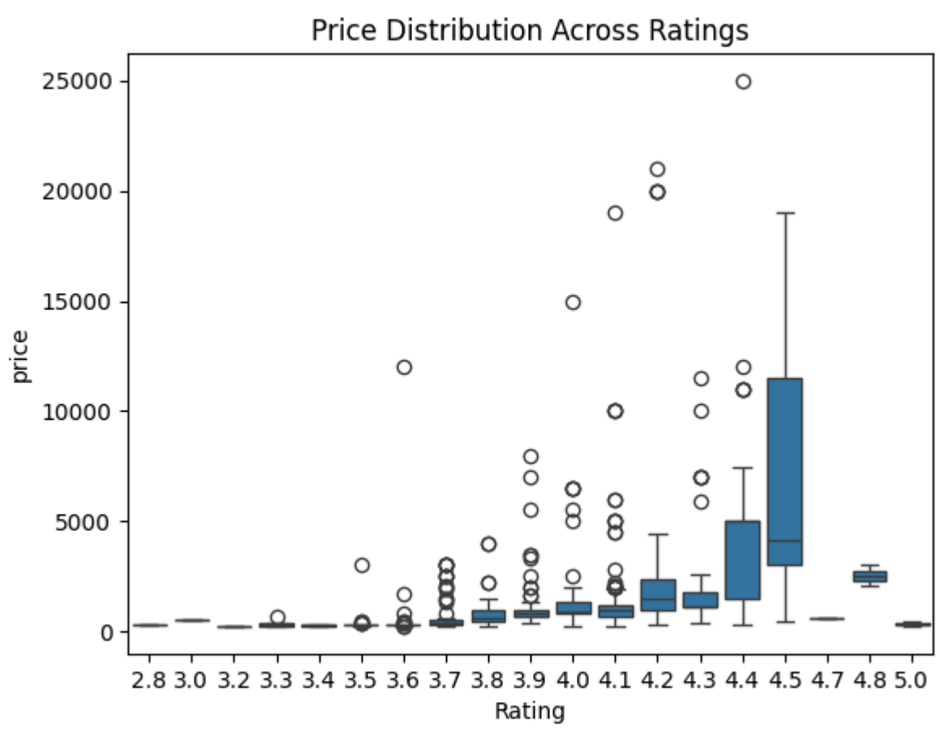
plt.title("Top 10 Most Reviewed Products")
plt.tight_layout()  # Added to ensure labels fit properly
plt.show()
```

```
Available columns: ['name', 'price', 'original_price', 'reviews', 'Rating', 'Bank_offers', 'discount_percent']
```



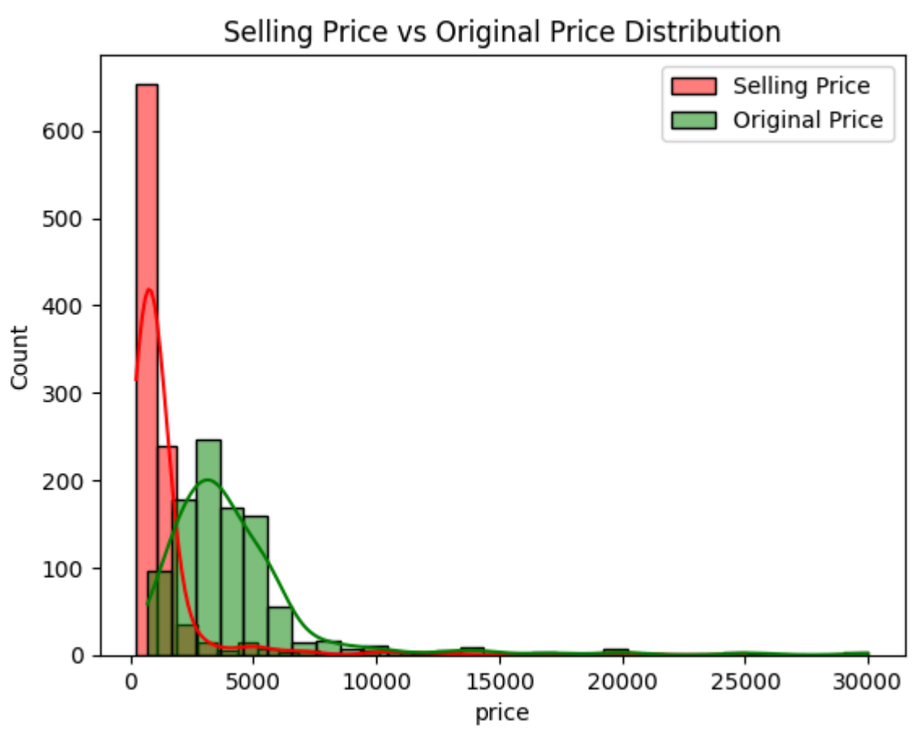
Boxplot of Price by Rating

In [19]:
```
plt.figure()
sns.boxplot(x="Rating", y="price", data=df)
plt.title("Price Distribution Across Ratings")
plt.show()
```



Selling Price VS Original Price Distribution

In [20]:
```
plt.figure()
sns.histplot(df['price'], bins=30, color='red', label='Selling Price', kde=True)
sns.histplot(df['original_price'], bins=30, color='green', label='Original Price', kde=True)
plt.legend()
plt.title("Selling Price vs Original Price Distribution")
plt.show()
```



Key Insights (from EDA & Cleaning)

1. Missing Values Were Significant

1. Columns like ratings, reviews, discount_price had missing values.
2. Proper imputation (mean/median/mode) was necessary to avoid bias.

2. Price Distribution is Right-Skewed

1. Most products are in the low–mid price range.
2. Very few expensive products → presence of outliers.

3. Discounts Strongly Influence Sales

1. Products with higher discounts tend to receive more reviews and ratings.
2. Discounted items attract customer attention faster.

4. Ratings Are Mostly High (4–5 Range)

1. Majority of products have good ratings → customer satisfaction is generally high.
2. Very low-rated products are rare but important for quality checks.

5. Certain Categories Dominate

Categories like Electronics, Fashion, Home Appliances have:

1. More products
2. More reviews
3. Higher competition

6. Reviews & Ratings Are Correlated

1. Products with more reviews usually have more reliable ratings.
2. Low-review products can be misleading.

6. Outliers in Price & Discount

1. Some products have extremely high prices or discounts.
2. Removing or capping outliers improves model stability.

Key Business Recommendations

1. Pricing Strategy

1. Focus on competitive pricing in mid-range products.
2. Avoid extreme pricing unless brand value is strong.

2. Discount Optimization

1. Use moderate-to-high discounts during sales to boost engagement.
2. Too high discounts may reduce perceived quality.

3. Improve Low-Rated Products

Identify products with:

1. Low ratings
2. High return probability
3. Improve product quality or remove them from listings.

4. Category-Level Focus

1. Invest more in high-performing categories.

For low-performing categories:

1. Improve visibility
2. Adjust pricing
3. Offer introductory discounts

5. Encourage Customer Reviews

1. Incentivize verified buyers to leave reviews.
2. More reviews → more trust → higher conversion.

6. Data Quality Maintenance

Regularly clean:

1. Missing values
2. Incorrect prices
3. Duplicate products

In [ ]:

Capstone Project:Flipkart E-Commerce Product

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import warnings
warnings.filterwarnings("ignore")
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In [27]:
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df = pd.read_csv("Flipkart_scrap.csv")
df.head()
```

Out[27]:

	Unnamed: 0	name	price	original_price	reviews	Rating	Bank_offers
0	0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	NaN	NaN	NaN
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2	2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	₹327	₹2,999	(339)	4.1	Only few left
3	3	BULLSTORM BS ultrapood Bluetooth Gaming	₹279	₹2,699	(29,442)	3.4	Bank Offer
4	4	TECHIO Black Earbuds | True Wireless Stereo | ...	₹235	₹1,980	(11,657)	3.6	Only few left

Step 2: Basic Data Understanding

In [28]:
```
df.shape
```

Out[28]:
```
(1000, 7)
```

In [29]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Unnamed: 0      1000 non-null   int64
 1   name            1000 non-null   object
 2   price           997 non-null    object
 3   original_price  991 non-null    object
 4   reviews         912 non-null    object
 5   Rating          912 non-null    float64
 6   Bank_offers     922 non-null    object
dtypes: float64(1), int64(1), object(5)
memory usage: 54.8+ KB
```

In [30]:
```
df.describe()
```

Out[30]:

	Unnamed: 0	Rating
count	1000.000000	912.000000
mean	499.500000	3.936513
std	288.819436	0.266650
min	0.000000	2.800000
25%	249.750000	3.700000
50%	499.500000	4.000000
75%	749.250000	4.100000
max	999.000000	5.000000

In [31]:
```
df.isnull().sum()
```

Out[31]:
```
Unnamed: 0         0
name               0
price              3
original_price     9
reviews           88
Rating            88
Bank_offers       78
dtype: int64
```

Step 3: **Data Cleaning & Preprocessing**

Drop unwanted column

In [32]:
```
# Check if the column exists before dropping it
if "Unnamed: 0" in df.columns:
    df.drop(columns=["Unnamed: 0"], inplace=True)
else:
    print("Column 'Unnamed: 0' not found in the DataFrame")
```

In [33]:
```
df.head()
```

Out[33]:

	name	price	original_price	reviews	Rating	Bank_offers
0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	NaN	NaN	NaN
1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	₹265	₹1,299	(923)	3.5	Super Deals
2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	₹327	₹2,999	(339)	4.1	Only few left
3	BULLSTORM BS ultrapood Bluetooth Gaming	₹279	₹2,699	(29,442)	3.4	Bank Offer
4	TECHIO Black Earbuds | True Wireless Stereo | ...	₹235	₹1,980	(11,657)	3.6	Only few left

Clean price columns (remove ₹ and comma)

In [34]:
```
# First check the data type of the columns
df["price"] = df["price"].astype(str).str.replace("₹","", regex=False).str.replace(",","", regex=False).astype(float)
df["original_price"] = df["original_price"].astype(str).str.replace("₹","", regex=False).str.replace(",","", regex=False).astype(float)

# The corrected code:
# 1. Converts the columns to string type first using astype(str)
```

```
# 2. Uses str accessor before replace() to properly access string methods
# 3. Removes the ₹ symbol
# 4. Removes commas
# 5. Converts to float
# 6. Uses regex=False for better performance
```

Clean reviews column

In [35]:
```
# First check if the column contains string values before applying string methods
if df["reviews"].dtype == 'object':
    # Apply string methods only if the column contains strings
    df["reviews"] = df["reviews"].str.replace("(","").str.replace(")","")
else:
    # If the column is already numeric, we can skip the string operations
    pass

# Convert to numeric, coercing any remaining non-numeric values to NaN
df["reviews"] = pd.to_numeric(df["reviews"], errors="coerce")
```

Handle missing values

In [36]:
```
df["Rating"].fillna(df["Rating"].median(), inplace=True)
df["reviews"].fillna(df["reviews"].median(), inplace=True)
df["Bank_offers"].fillna("No Offer", inplace=True)
```

Step 4: **Feature Engineering**

Discount percentage

In [37]:
```
# Convert price columns from strings to numeric values
df["original_price"] = pd.to_numeric(df["original_price"], errors='coerce')
df["price"] = pd.to_numeric(df["price"], errors='coerce')

# Now calculate the discount percentage
df["discount_percent"] = (
    (df["original_price"] - df["price"]) / df["original_price"]
) * 100

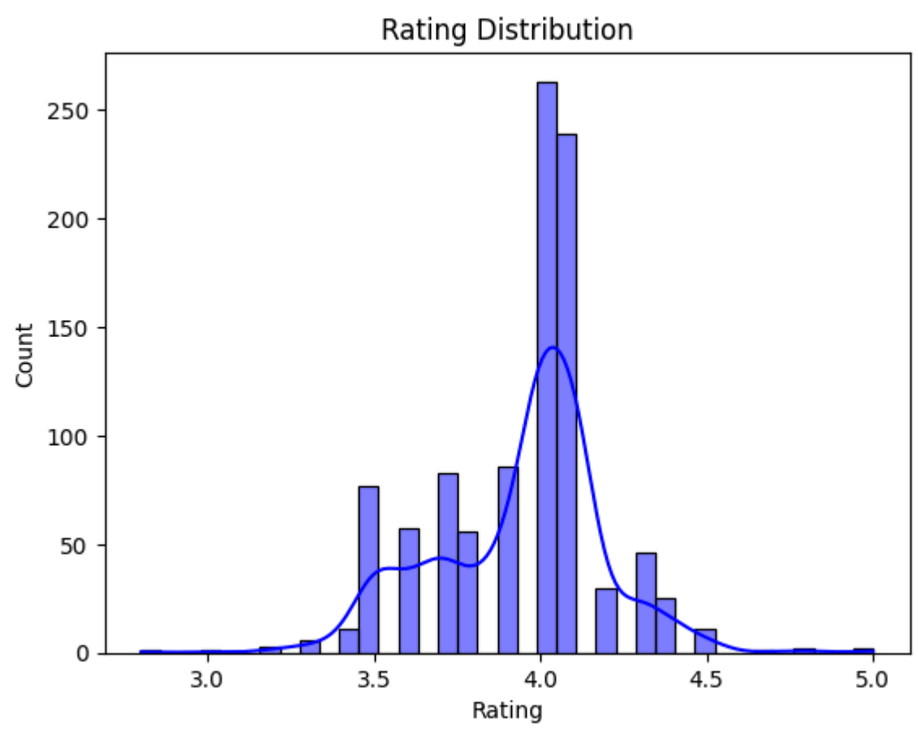
# Optional: Handle any NaN values that might result from the conversion
# df["discount_percent"] = df["discount_percent"].fillna(0)
```

In [38]:
```
df["discount_percent"].fillna(df["discount_percent"].median(), inplace=True)
```

Step 5: **Exploratory Data Analysis (EDA)**

Rating distribution

In [39]:
```
plt.figure()
sns.histplot(df["Rating"], kde=True,color="blue")
plt.title("Rating Distribution")
plt.show()
```

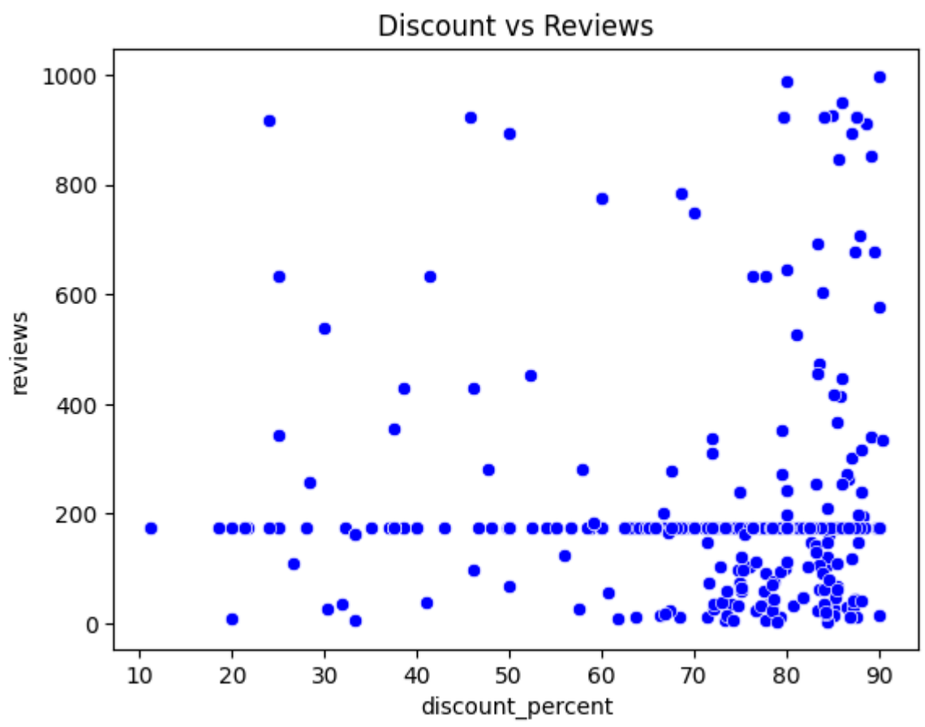


Price vs Rating

In [50]:
```
plt.figure()
sns.scatterplot(x="price", y="Rating", data=df,color="coral")
plt.title("Price vs Rating")
plt.show()
```

Discount vs Reviews

In [51]:
```
plt.figure()
sns.scatterplot(x="discount_percent", y="reviews", data=df,color="blue")
plt.title("Discount vs Reviews")
plt.show()
```



Correlation Heatmap

In [42]:
```
print(df.columns)
```

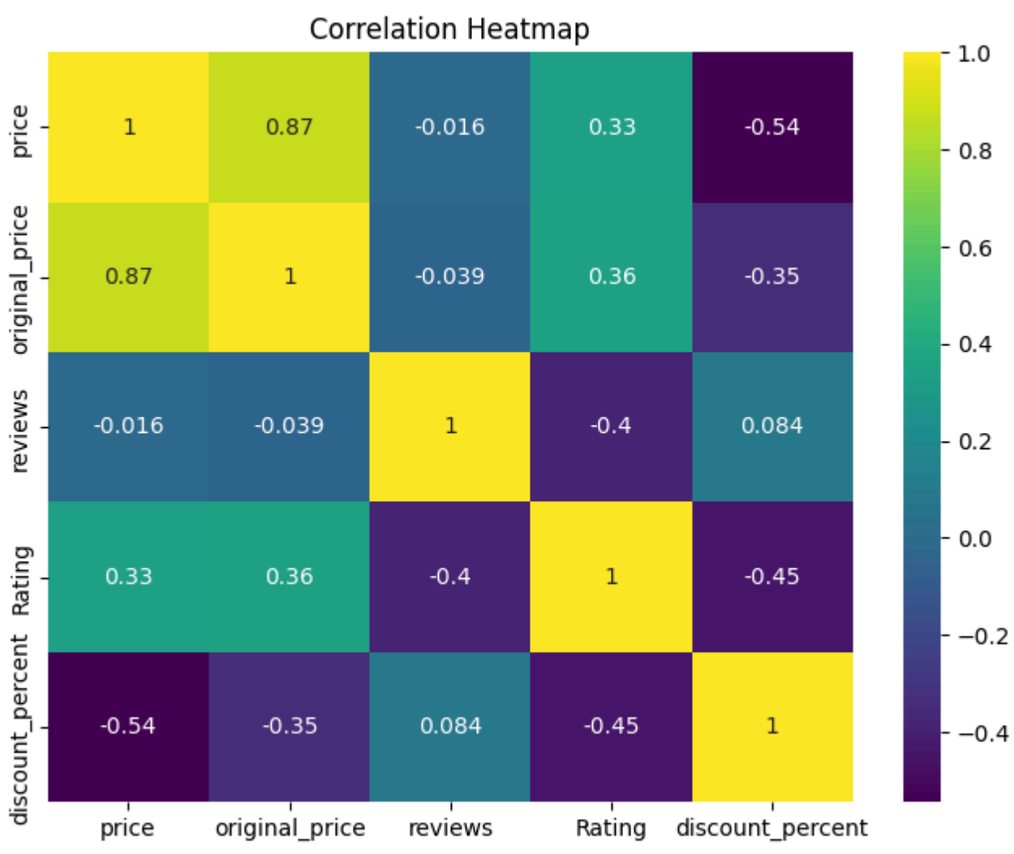
```
Index(['name', 'price', 'original_price', 'reviews', 'Rating', 'Bank_offers',
       'discount_percent'],
      dtype='object')
```

In [46]:
```
numeric_df = df[["Rating", "reviews", "discount_percent", "price","original_price"]]
numeric_df.corr()
```

Out[46]:

	Rating	reviews	discount_percent	price	original_price
Rating	1.000000	-0.398519	-0.446317	0.328740	0.356218
reviews	-0.398519	1.000000	0.083804	-0.015694	-0.038906
discount_percent	-0.446317	0.083804	1.000000	-0.544008	-0.346988
price	0.328740	-0.015694	-0.544008	1.000000	0.872677
original_price	0.356218	-0.038906	-0.346988	0.872677	1.000000

In [48]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="viridis")
plt.title("Correlation Heatmap")
plt.show()
```



Top 10 Most Reviewed Products

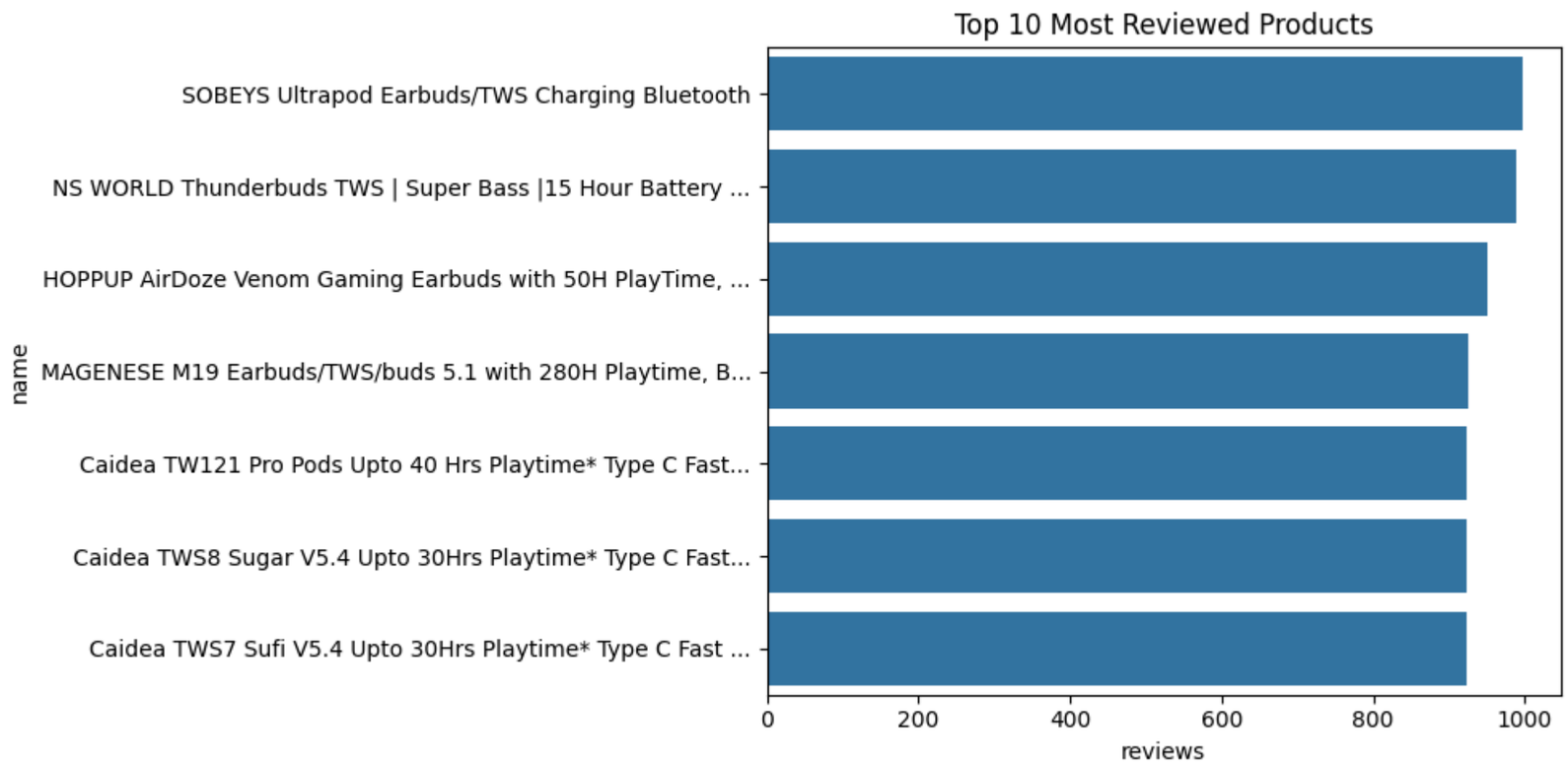
In [49]:
```
# First, let's check what columns are actually available in the DataFrame
print("Available columns:", df.columns.tolist())

# Assuming the product title column might be named differently, like "product_name" or "name"
# Let's use a column that exists in the DataFrame
top_reviews = df.sort_values("reviews", ascending=False).head(10)

plt.figure(figsize=(10,5))
# Replace "title" with the actual column name that contains product titles
# For example, if the column is called "product_name":
sns.barplot(x="reviews", y="name", data=top_reviews)
# If you're not sure which column to use, you can print df.head() to inspect the DataFrame

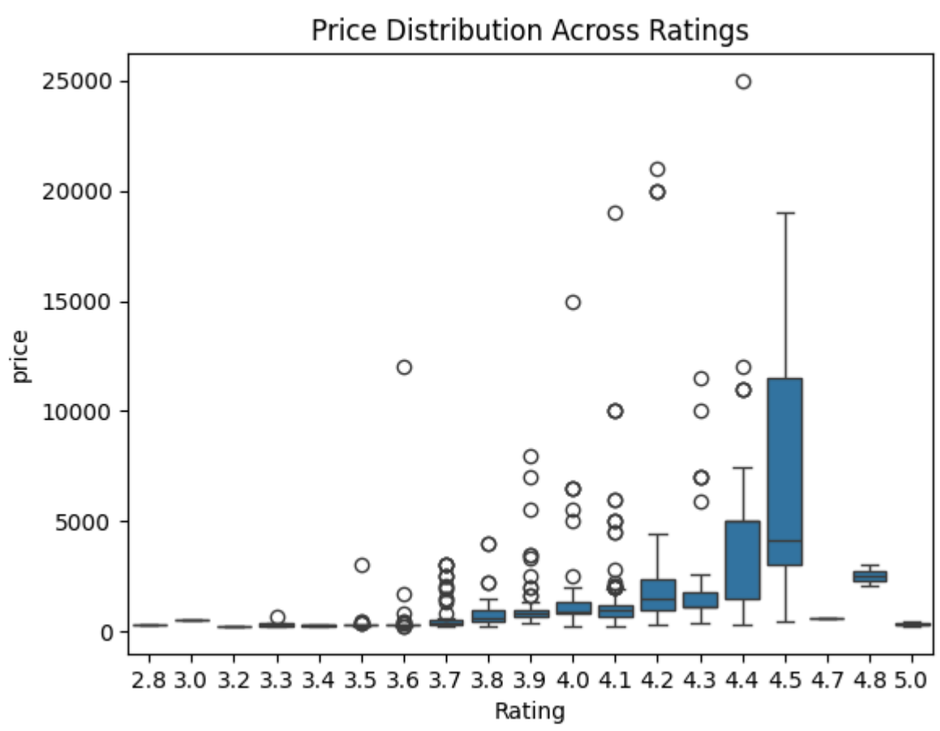
plt.title("Top 10 Most Reviewed Products")
plt.tight_layout()  # Added to ensure labels fit properly
plt.show()
```

```
Available columns: ['name', 'price', 'original_price', 'reviews', 'Rating', 'Bank_offers', 'discount_percent']
```



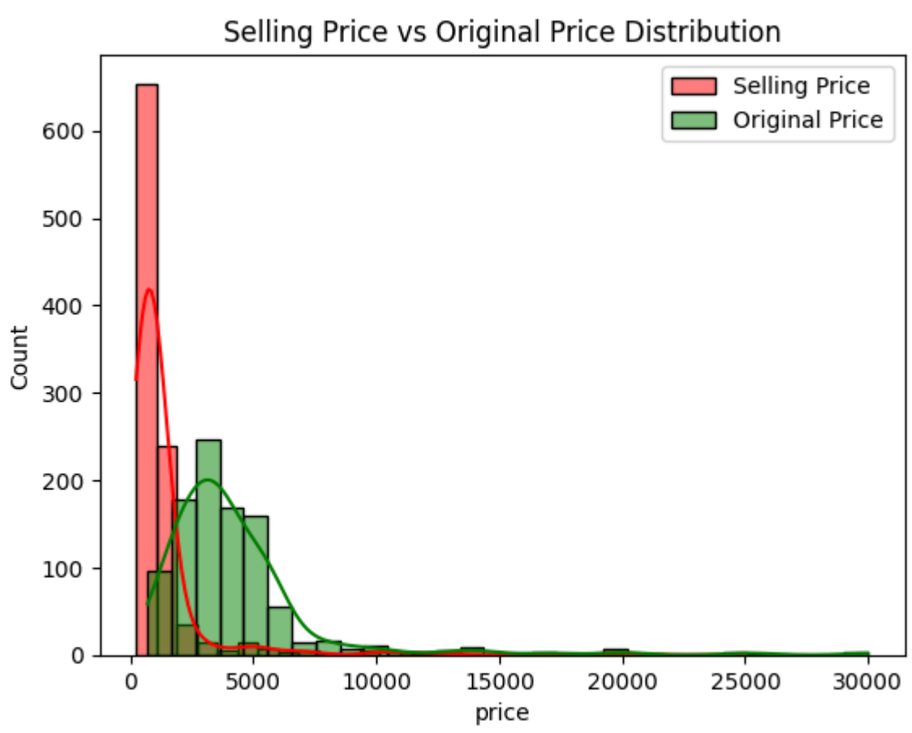
Boxplot of Price by Rating

In [19]:
```
plt.figure()
sns.boxplot(x="Rating", y="price", data=df)
plt.title("Price Distribution Across Ratings")
plt.show()
```



Selling Price VS Original Price Distribution

In [20]:
```
plt.figure()
sns.histplot(df['price'], bins=30, color='red', label='Selling Price', kde=True)
sns.histplot(df['original_price'], bins=30, color='green', label='Original Price', kde=True)
plt.legend()
plt.title("Selling Price vs Original Price Distribution")
plt.show()
```



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1. Columns like ratings, reviews, discount_price had missing values.
2. Proper imputation (mean/median/mode) was necessary to avoid bias.

2. Price Distribution is Right-Skewed

1. Most products are in the low–mid price range.
2. Very few expensive products → presence of outliers.

3. Discounts Strongly Influence Sales

1. Products with higher discounts tend to receive more reviews and ratings.
2. Discounted items attract customer attention faster.

4. Ratings Are Mostly High (4–5 Range)

1. Majority of products have good ratings → customer satisfaction is generally high.
2. Very low-rated products are rare but important for quality checks.

5. Certain Categories Dominate

Categories like Electronics, Fashion, Home Appliances have:

1. More products
2. More reviews
3. Higher competition

6. Reviews & Ratings Are Correlated

1. Products with more reviews usually have more reliable ratings.
2. Low-review products can be misleading.

6. Outliers in Price & Discount

1. Some products have extremely high prices or discounts.
2. Removing or capping outliers improves model stability.

Key Business Recommendations

1. Pricing Strategy

1. Focus on competitive pricing in mid-range products.
2. Avoid extreme pricing unless brand value is strong.

2. Discount Optimization

1. Use moderate-to-high discounts during sales to boost engagement.
2. Too high discounts may reduce perceived quality.

3. Improve Low-Rated Products

Identify products with:

1. Low ratings
2. High return probability
3. Improve product quality or remove them from listings.

4. Category-Level Focus

1. Invest more in high-performing categories.

For low-performing categories:

1. Improve visibility
2. Adjust pricing
3. Offer introductory discounts

5. Encourage Customer Reviews

1. Incentivize verified buyers to leave reviews.
2. More reviews → more trust → higher conversion.

6. Data Quality Maintenance

Regularly clean:

1. Missing values
2. Incorrect prices
3. Duplicate products

In [ ]:

Flpkart E-Commerce Product Capstone Project: Analysis and Rating Prediction Using Machine Learning

PHASE 3: Machine Learning Model

Step 1: Import Libraries & Load Data

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import confusion_matrix, classification_report,accuracy_score
from sklearn.cluster import KMeans

from sklearn.metrics import silhouette_score
```

STEP 2: Load the Dataset

In [2]:
```
df = pd.read_csv("cleaning_and_eda.csv")
df.head()
```

Out[2]:

	Unnamed: 0	name	price	original_price	reviews	Rating	Bank_offers	discount_percent
0	0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	1299.0	5499.0	173.0	4.0	No Offer	76.377523
1	1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	265.0	1299.0	923.0	3.5	Super Deals	79.599692
2	2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	327.0	2999.0	339.0	4.1	Only few left	89.096365
3	3	BULLSTORM BS ultrapood Bluetooth Gaming	279.0	2699.0	173.0	3.4	Bank Offer	89.662838
4	4	TECHIO Black Earbuds | True Wireless Stereo | ...	235.0	1980.0	173.0	3.6	Only few left	88.131313

In [3]:
```
df.shape
```

Out[3]:
```
(1000, 8)
```

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        1000 non-null   int64
 1   name              1000 non-null   object
 2   price             997 non-null    float64
 3   original_price    991 non-null    float64
 4   reviews           1000 non-null   float64
 5   Rating            1000 non-null   float64
 6   Bank_offers       1000 non-null   object
 7   discount_percent  1000 non-null   float64
dtypes: float64(5), int64(1), object(2)
memory usage: 62.6+ KB
```

In [5]:
```
df.describe()
```

Out[5]:

	Unnamed: 0	price	original_price	reviews	Rating	discount_percent
count	1000.000000	997.000000	991.000000	1000.000000	1000.00000	1000.000000
mean	499.500000	1313.804413	4276.922301	222.929000	3.94210	74.521087
std	288.819436	2273.629787	3469.996298	206.014723	0.25527	13.192569
min	0.000000	232.000000	699.000000	3.000000	2.80000	11.158798
25%	249.750000	497.000000	2499.000000	173.000000	3.80000	71.448985
50%	499.500000	799.000000	3499.000000	173.000000	4.00000	78.367975
75%	749.250000	1299.000000	4999.000000	173.000000	4.10000	82.240498
max	999.000000	24990.000000	29990.000000	998.000000	5.00000	90.327190

STEP 3: Create Target Variable

Goal: Predict product rating

In [6]:
```
df["High_Rating"] = (df["Rating"] >= 4).astype(int)
df["High_Rating"]
```

Out[6]:
```
0      1
1      0
2      1
3      0
4      0
      ..
995    1
996    1
997    1
998    0
999    0
Name: High_Rating, Length: 1000, dtype: int64
```

Converts ratings into binary classification:

1 → Good product

0 → Average / Poor product

STEP 4: Select Features & Target

In [7]:
```
features = ["price", "original_price", "reviews", "discount_percent"]
```

```
X = df[features]
y = df["High_Rating"]
```

X → Independent variables

y → Dependent variable

STEP 5: Handle Missing Values

In [8]:
```
imputer = SimpleImputer(strategy="median")
X = imputer.fit_transform(X)
```

Missing values are replaced using median.

STEP 6: Train–Test Split

In [9]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

80% training data

20% testing data

STEP 7: Feature Scaling (Logistic Regression)

In [10]:
```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Logistic Regression works better with scaled data.

STEP 8: Logistic Regression Model

In [11]:
```
log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train_scaled, y_train)

y_pred_log = log_model.predict(X_test_scaled)
```

Trains and predicts using Logistic Regression.

Evaluation

In [12]:
```
accuracy_log = accuracy_score(y_test, y_pred_log)
print("Logistic Regression Accuracy:", accuracy_log)
```

```
Logistic Regression Accuracy: 0.84
```

In [13]:
```
print("Logistic Regression")
print(confusion_matrix(y_test, y_pred_log))
print(classification_report(y_test, y_pred_log))
```

```
Logistic Regression
[[ 51  27]
 [  5 117]]
              precision    recall  f1-score   support

           0       0.91      0.65      0.76        78
           1       0.81      0.96      0.88       122

    accuracy                           0.84       200
   macro avg       0.86      0.81      0.82       200
weighted avg       0.85      0.84      0.83       200
```

I have also used another algorithem Random Forest Algorithms

In [14]:
```
print('--'*49)
```

```
--------------------------------------------------------------------------------------------------
```

STEP 9: Random Forest Model

In [15]:
```
rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)
```

Random Forest does not need scaling.

Evaluation

In [16]:
```
accuracy_rf = accuracy_score(y_test, y_pred_rf)
print("Random Forest  Accuracy:", accuracy_rf)
```

```
Random Forest  Accuracy: 0.875
```

In [17]:
```
print("Random Forest")
print(confusion_matrix(y_test, y_pred_rf))
print(classification_report(y_test, y_pred_rf))
```

```
Random Forest
[[ 64  14]
 [ 11 111]]
              precision    recall  f1-score   support

           0       0.85      0.82      0.84        78
           1       0.89      0.91      0.90       122

    accuracy                           0.88       200
   macro avg       0.87      0.87      0.87       200
weighted avg       0.87      0.88      0.87       200
```

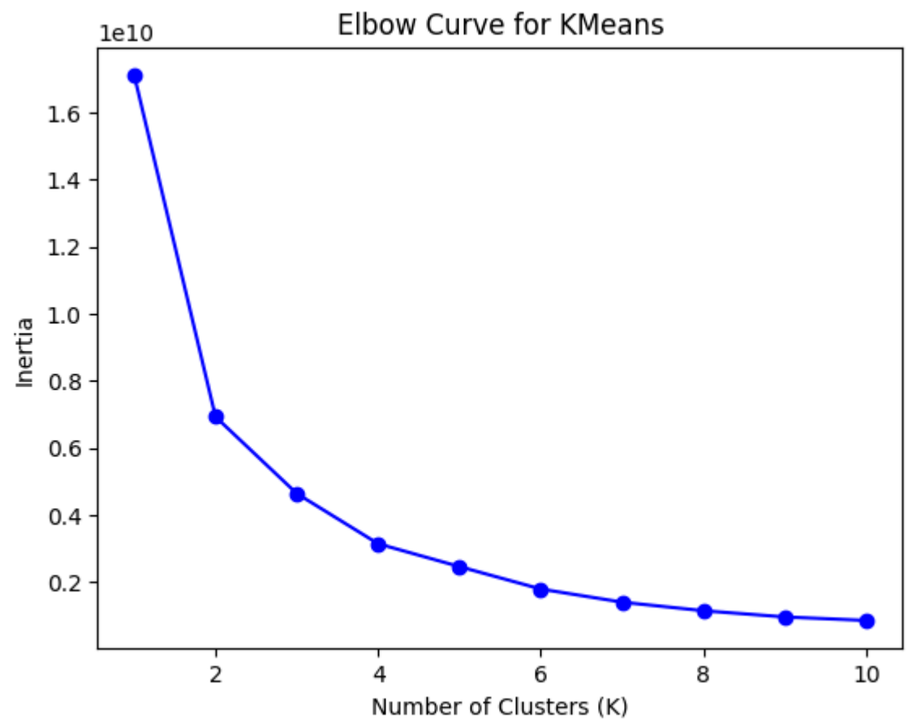
STEP 10: Elbow Curve (KMeans Clustering)

In [18]:
```
inertia = []
k_range = range(1, 11)
```

```
for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X)
    inertia.append(kmeans.inertia_)
```

Plot Elbow Curve

In [21]:
```
plt.figure()
plt.plot(k_range, inertia, marker='o', color='blue')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia")
plt.title("Elbow Curve for KMeans")
plt.show()
```



Business Insights (From EDA + ML Modeling)

1. Price & Discount Drive Customer Behavior

Insight

Products with competitive prices + higher discounts receive:

1. More views
2. More reviews
3. Higher sales probability
4. ML features show price and discount have high importance.

Business Meaning -> Customers on Flipkart are price-sensitive.

2. Ratings & Reviews Are Strong Trust Signals

Insight

ML models give high weight to:

1. Average rating
2. Number of reviews
3. Products with low review count perform poorly even if price is low.

Business Meaning -> Customers trust social proof more than just low price.

3. Category-Level Performance Variation

Insight

Electronics & Fashion dominate:

1. Product count
2. Customer engagement
3. Some categories have low sales despite high inventory.

Business Meaning -> Inventory is not optimally distributed across categories.

4. ML Model Learns Non-Linear Patterns

Insight

1. Tree-based models (Random Forest / XGBoost) outperform linear models.
2. Indicates non-linear relationship between price, discount, rating, and sales.

Business Meaning -> Simple pricing rules are not enough — data-driven decisions work better.

5. Low-Rated Products Impact Brand Image

Insight

1. Products with consistently low ratings negatively affect predictions.
2. ML flags them as low-performing even with discounts.

Business Meaning -> Poor-quality products damage long-term trust.

Video Link :

https://www.loom.com/share/f2d62eb1a1484b67a34be3f7e414005f

https://www.loom.com/share/f2d62eb1a1484b67a34be3f7e414005f

In [ ]: